

7. Software Architecture and Detailed Design

This section defines how TriageAI will be built. Section 7.1 presents the system architecture, the deployment design, and the detailed design of the triage pipeline, API, database, and source code. Section 7.2 records the main architecture decisions with their rationale and the design principles applied. Section 7.3 presents the user-interface wireframes and prototype screens. The design realizes the requirements of the SRS (Version 1.1) and the analysis models of PD6; references such as FR-23, ADR-08, or W-04 point to requirements, decisions in Section 7.2, and screens in Section 7.3.

7.1 System Architecture

7.1.1 Architectural Style and Overview

TriageAI uses a three-tier, layered architecture implemented as a modular monolith (ADR-01). The client tier is a single-page web application; the application tier is one backend service divided into clearly separated layers and modules; and the data tier is a single PostgreSQL database that also stores the vector index (ADR-04). Within the backend, the triage pipeline follows a pipe-and-filter design in which six stages transform the case step by step, the external AI services are reached only through adapters (ADR-06, ADR-07), and all persistence goes through a repository-based data access layer. Figure 7.1 shows the logical architecture.

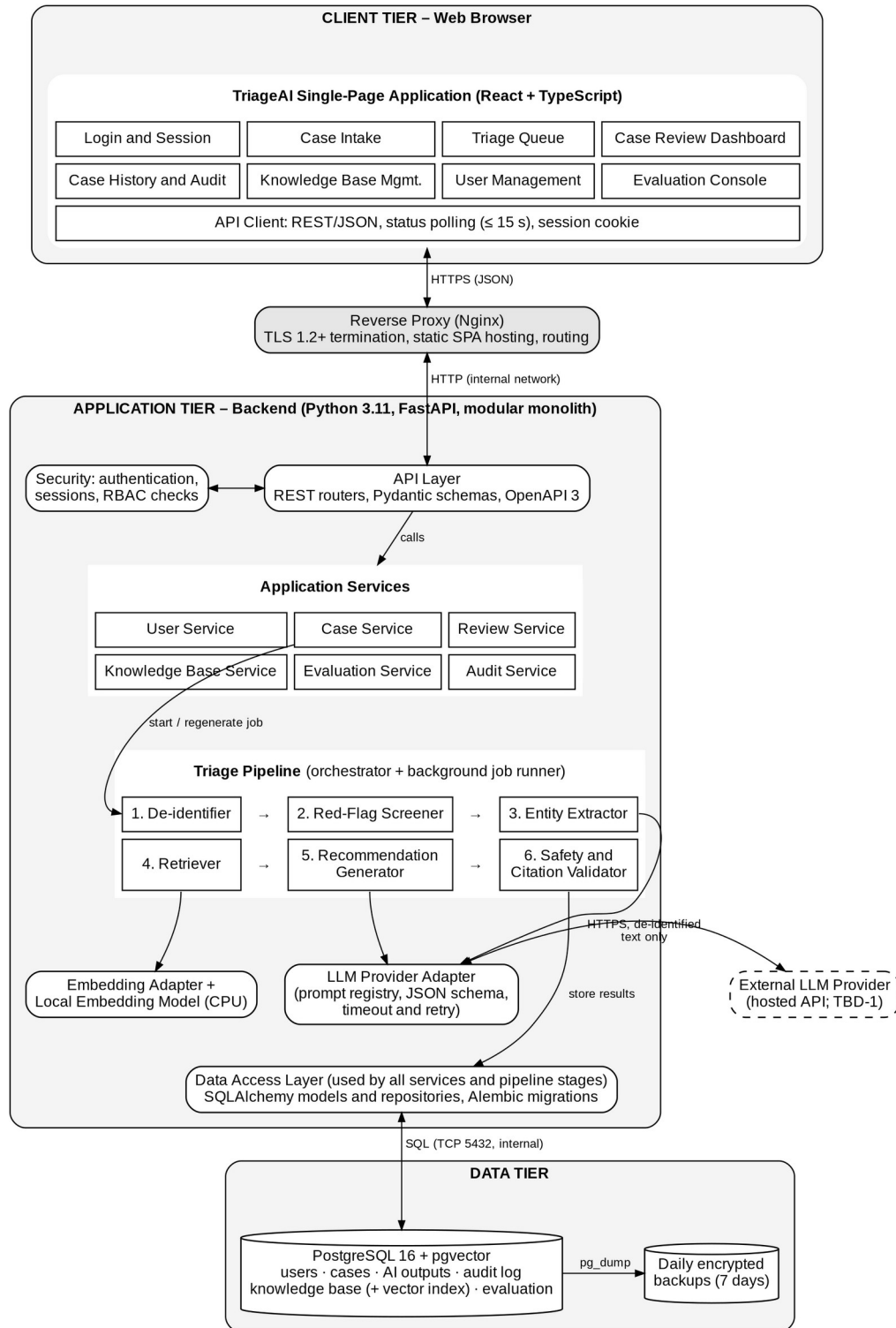


Figure 7.1: TriageAI Logical Architecture

7.1.2 Component Descriptions

The components of Figure 7.1, their responsibilities, and the PD6 processes they realize are summarized below.

COMPONENT	RESPONSIBILITY	TECHNOLOGY	REALIZES
Single-Page Application	Renders all screens (Section 7.3), validates input on the client, and polls case status.	React 18, TypeScript, Vite	All user interfaces (IR-01–IR-08)
Reverse Proxy	Terminates TLS, serves the static SPA build, and forwards /api requests to the backend.	Nginx	IR-16, SR-06
API Layer	Exposes the REST endpoints (Section 7.1.5), validates requests and responses, and publishes the OpenAPI document.	FastAPI, Pydantic	IR-11, IR-15
Security	Authenticates users, manages sessions and lockout, and enforces role checks on every endpoint.	FastAPI dependencies, Argon2id, signed JWT cookies	Process 1.0; SR-01–SR-05
User Service	Creates, edits, deactivates, and unlocks accounts; assigns roles and permissions.	Python module	Process 1.0; FR-58–FR-63
Case Service	Creates cases, stores descriptions verbatim, and starts pipeline jobs.	Python module	Process 2.0; FR-01–FR-08
Triage Pipeline	Runs the six pipeline stages as a background job (Section 7.1.4) and records every output and configuration.	Python modules, job runner	Processes 3.0 and 4.0; FR-09–FR-28
Review Service	Builds the Triage Queue and review data; records decisions, notes, corrections, and closures.	Python module	Process 5.0; FR-29–FR-40, FR-48
Knowledge Base Service	Manages the entry workflow, approval, chunking, embedding, and version publishing.	Python module	Process 6.0; FR-49–FR-56
Evaluation Service	Imports vignettes, runs evaluations through the pipeline, computes metrics, and exports results.	Python module, scikit-learn	Process 7.0; FR-64–FR-70
Audit Service	Writes append-only audit entries for case and security events and answers audit queries.	Python module	FR-41–FR-47, SR-12
LLM Provider Adapter	Loads versioned prompt templates, calls the LLM with a JSON schema, applies timeout and retry, and hides provider details.	HTTP client (httpx)	IR-12, IR-13, IR-18, IR-19
Embedding Adapter	Produces embeddings for knowledge base chunks and retrieval queries using a locally hosted model.	sentence-transformers (CPU)	ADR-07; TBD-2

COMPONENT	RESPONSIBILITY	TECHNOLOGY	REALIZES
Data Access Layer	Maps domain classes to tables and provides repositories for each aggregate; manages schema migrations.	SQLAlchemy 2, Alembic	DR-01–DR-08
Database	Stores relational data and the vector index in one PostgreSQL instance.	PostgreSQL 16, pgvector	Data stores D1–D6
Backup Job	Creates a daily encrypted dump of the database and keeps the last seven.	cron, pg_dump	SR-15, DR-07

7.1.3 Deployment Diagram

Figure 7.2 shows the physical deployment for the development and evaluation environments (ADR-13). All server components run as Docker containers on a single Linux virtual machine managed with Docker Compose. Only the reverse proxy is reachable from the network; the backend and the database communicate over an internal Docker network. The same Compose file runs on team members’ laptops for development, so the evaluation environment differs only in its configuration values and TLS certificate.

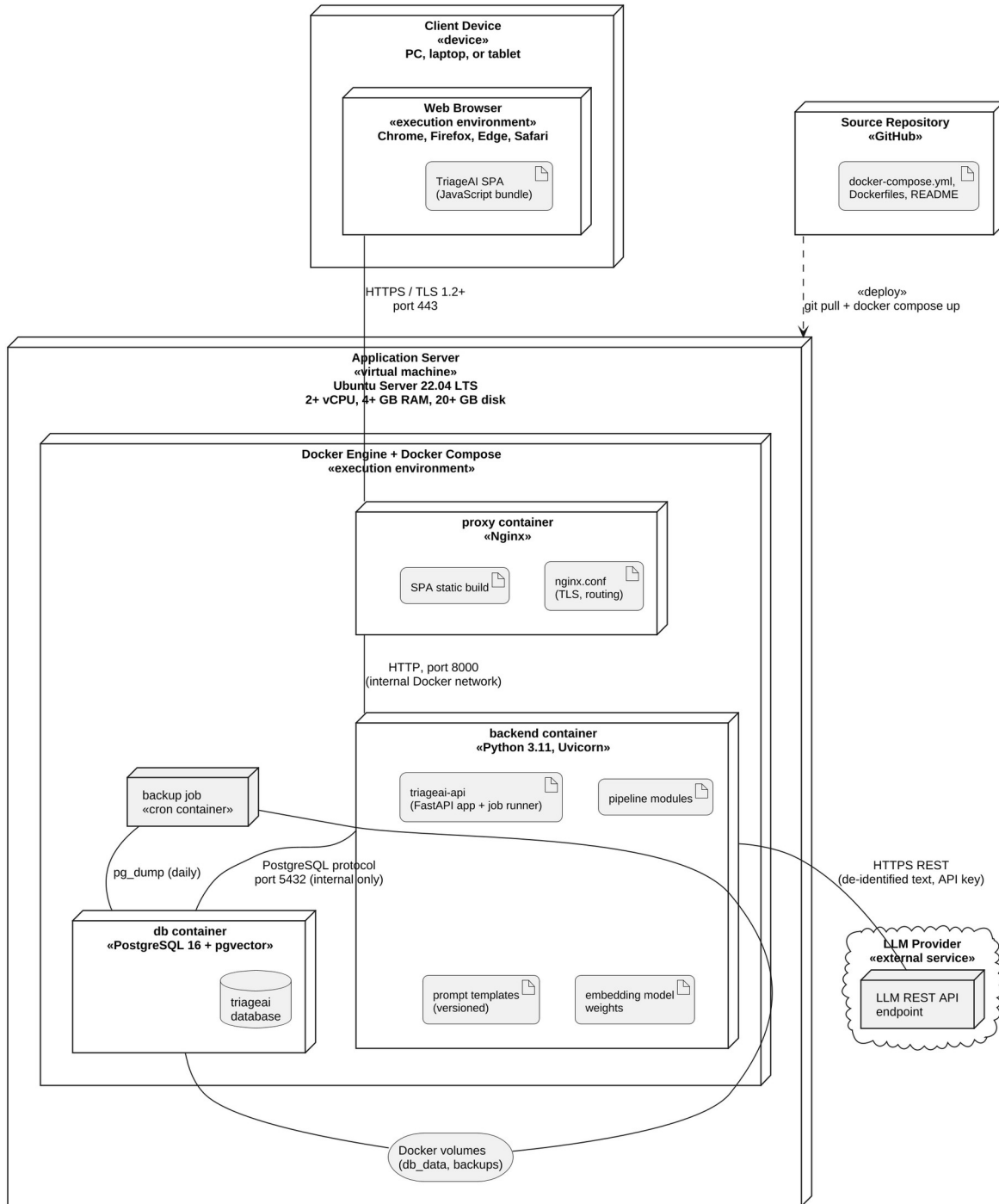


Figure 7.2: TriageAI Deployment Diagram

NODE / ARTIFACT	DESCRIPTION	SPECIFICATION
Client Device	Staff computer or tablet running a supported browser.	Chrome, Firefox, Edge, Safari (current and previous versions); ≥ 1280 × 720 or ≥ 768 px tablet

NODE / ARTIFACT	DESCRIPTION	SPECIFICATION
Application Server	Virtual machine hosting all containers (hosting provider TBD-3).	Ubuntu Server 22.04 LTS; ≥ 2 vCPU, 4 GB RAM (8 GB recommended for the embedding model), 20 GB disk
proxy container	Nginx serving the SPA and proxying the API; only public entry point.	Ports 80 $\rightarrow$ 443 redirect, 443 (TLS 1.2+)
backend container	FastAPI application with the job runner, pipeline, prompts, and embedding model.	Python 3.11, Uvicorn, port 8000 (internal)
db container	PostgreSQL with pgvector; data kept on a Docker volume.	PostgreSQL 16, port 5432 (internal only)
backup job	Daily pg_dump to the backups volume; encrypted; 7-day retention.	cron container
LLM Provider	External hosted model reached over HTTPS with a server-side API key.	Selected in TBD-1; accessed only with de-identified text
Source Repository	Holds the code, Dockerfiles, docker-compose.yml, and README used for deployment.	GitHub; accessible to the professors

7.1.4 Detailed Design of the Triage Pipeline

The design class diagram in Figure 7.3 refines Processes 3.0 and 4.0 of PD6 into implementation classes. TriagePipeline orchestrates six single-purpose stage classes. The Entity Extractor and the Recommendation Generator depend on the LLMProvider interface, and the Retriever depends on the EmbeddingProvider interface, so a provider can be replaced or mocked in tests without changing the pipeline (dependency inversion). Repositories and the Audit Service are also accessed through interfaces.

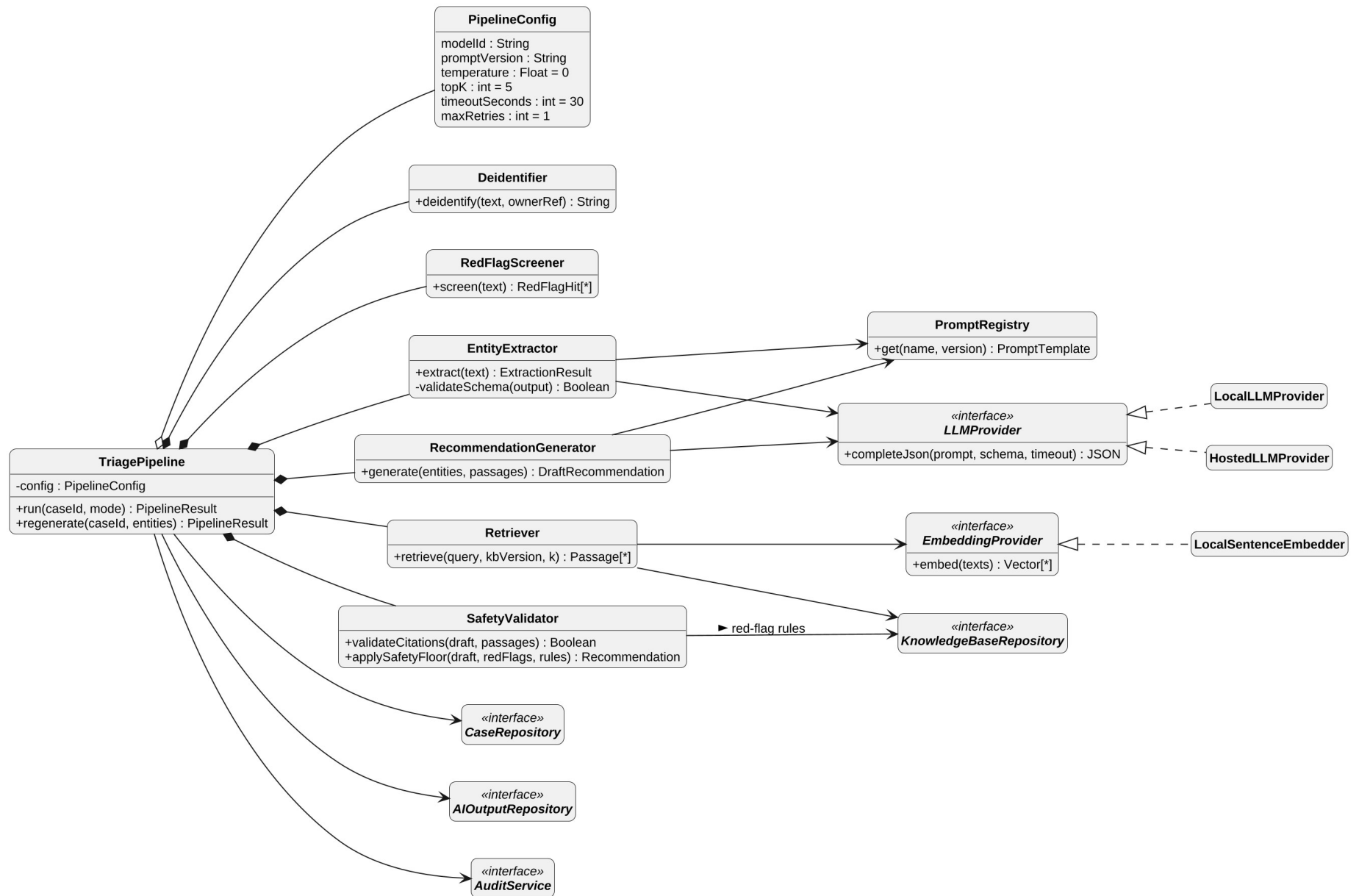


Figure 7.3: Design Class Diagram of the Triage Pipeline

The behavior of each stage, including its failure handling, is specified below. Every stage records its output, duration, and configuration so that the evaluation can report stage latencies (FR-68) and results can be reproduced (NFR-23).

STAGE	INPUT → OUTPUT	DESIGN AND FAILURE HANDLING	REQUIREMENTS
1. De-identifier	Description, owner reference → de-identified text	Removes the stored owner name and contact values and pattern-matched phone numbers, e-mail addresses, and street addresses. Runs locally; never fails open: if it raises an error, the job stops and the case is set to Manual Triage Required.	IR-20, SR-08, ADR-10
2. Red-Flag Screener	De-identified text, red-flag list → red-flag hits	Deterministic keyword and phrase matching in English against the approved red-flag list; alerts are written before any LLM call.	FR-12, NFR-05, ADR-09
3. Entity Extractor	De-identified text → validated extraction	Sends the extraction prompt with the JSON schema; validates the output with Pydantic; maps complaints to the fixed list; retries once on invalid output; timeout 30 s.	FR-09–FR-18, IR-13, IR-19
4. Retriever	Entities → top-k passages	Builds a query from the complaints and signs, embeds it locally, and runs a cosine-similarity search on the active KB version (default k = 5).	FR-19, NFR-02, ADR-07
5. Recommendation Generator	Entities, passages → draft recommendation	Sends the generation prompt with numbered passages and requires a JSON answer containing one category, a rationale, cited passage numbers, and a confidence value; temperature 0.	FR-20, FR-21, FR-24, ADR-14
6. Safety and Citation Validator	Draft, passages, red flags, rules → final recommendation	Rejects citations to passages that were not retrieved; applies the red-flag minimum category; selects the more urgent category on ties; labels low confidence when checks fail.	FR-22, FR-23, FR-25, NFR-12
Orchestrator and Job Runner	Case ID → stored results and status	Runs stages in order inside a background job; persists job state in the jobs table; on restart, re-queues jobs left in Processing; on any unrecoverable error, sets Manual Triage Required.	FR-06, FR-26, NFR-09, ADR-08

Prompt design. Prompt templates are stored as versioned files in the repository (for example, `extraction_v1.3` and `generation_v1.3`). System instructions and the owner’s text are placed in separate, clearly delimited sections, and the model is told to treat the owner’s text only as data (SR-

10). Both prompts require JSON output that is validated before use, and the prompt version is stored with every result.

7.1.5 API Design

The REST API finalizes the provisional endpoints used in the PD6 sequence diagrams. All paths are prefixed with /api/v1, exchange JSON in UTF-8, and require an authenticated session except for login. Standard status codes are used: 200/201 for success, 202 when a pipeline job has been accepted, 400/422 for validation errors, 401 when not authenticated, 403 when the role is not permitted, 404 when a record does not exist, 409 for version conflicts, and 503 when an AI service is unavailable. Role abbreviations: IS = Intake Staff, VR = Veterinary Reviewer, KBA = Veterinary Reviewer with knowledge base approval permission, AD = Administrator.

METHOD	ENDPOINT	PURPOSE	ROLES	USE CASE
POST	/auth/login	Authenticate and set the session cookie	All	UC-01
POST	/auth/logout	End the session	All	UC-01
POST	/auth/change-password	Change password (required after first login)	All	UC-01
GET	/auth/me	Return the current user and role	All	UC-01
POST	/cases	Create a case and start the pipeline (returns 202 and caseld)	IS, VR	UC-02
GET	/cases	Triage Queue and case search with filters	IS, VR, AD (view)	UC-04, UC-11
GET	/cases/{id}	Case review details	IS (view), VR	UC-05
GET	/cases/{id}/status	Pipeline status for polling	IS, VR	UC-02
PATCH	/cases/{id}/extraction	Save corrected extraction and regenerate (returns 202)	VR	UC-09
POST	/cases/{id}/decision	Record CONFIRM, ADJUST (category, reason), or MANUAL (category, note)	VR	UC-06–UC-08
POST	/cases/{id}/retry	Retry AI processing after a failure	VR	UC-08
POST	/cases/{id}/notes	Add a clinical note	VR	UC-05
POST	/cases/{id}/close	Close a case	VR	UC-10
GET	/cases/{id}/audit	Audit timeline of a case	IS, VR, AD	UC-11
GET / POST	/users	List users; create a user with a temporary password	AD	UC-12
PATCH	/users/{id}	Edit role, permission, or status; unlock; reset password	AD	UC-12

METHOD	ENDPOINT	PURPOSE	ROLES	USE CASE
GET / POST	/kb/entries	List entries by status; create a draft	AD, KBA	UC-13
PATCH	/kb/entries/{id}	Edit a draft entry	AD	UC-13
POST	/kb/entries/{id}/submit	Submit a draft for review	AD	UC-13
POST	/kb/entries/{id}/approve	Approve, embed, index, and publish a new version	KBA	UC-14
POST	/kb/entries/{id}/reject	Return an entry to Draft with comments	KBA	UC-14
POST	/kb/entries/{id}/retire	Retire an entry (new version)	AD	UC-13
GET	/kb/versions	List published knowledge base versions	AD, KBA	UC-13
POST	/eval/sets	Upload and validate a vignette file (CSV/JSON)	AD	UC-15
POST	/eval/runs	Start an evaluation run (returns 202)	AD	UC-15
GET	/eval/runs/{id}	Run status, metrics, and results	AD	UC-15
GET	/eval/runs/{id}/export	Export results (?format=csv json)	AD	UC-16
GET	/exports/cases	Export de-identified case and audit data (CSV)	AD	UC-16

7.1.6 Database Design

The PD6 domain model is mapped to the following PostgreSQL tables. Primary keys are UUIDs unless noted; all timestamps are stored in UTC (DR-03); enumerations are stored as PostgreSQL enum types that match the PD6 enumerations.

TABLE	KEY COLUMNS	CONSTRAINTS AND NOTES
users	id, email, password_hash, role, can_approve_kb, is_active, failed_login_count, locked_until	email unique; can_approve_kb only when role = VETERINARY_REVIEWER
cases	id, case_no, species, intake_channel, status, created_by → users, created_at, closed_at	case_no unique (C-0001...); index on (status, created_at) for the queue
signalments	case_id → cases, pet_name, age, sex, neutered, breed, weight_kg	one-to-one with cases
owner_descriptions	case_id → cases, text, submitted_at	application role has no UPDATE privilege (FR-05)
owner_references	case_id → cases, owner_name, contact_number	separate table; never read by the pipeline (DR-04)

TABLE	KEY COLUMNS	CONSTRAINTS AND NOTES
extraction_results	id, case_id or vignette_id, version, entities (jsonb), red_flags, missing_info, is_corrected, corrected_by	exactly one of case_id / vignette_id is set; unique (case_id, version)
extraction_complaints	extraction_id → extraction_results, complaint_code → presenting_complaints	junction table
recommendations	id, case_id or vignette_id, extraction_id, kb_version_id, version, category, rationale, confidence, safety_floor_applied, model_id, prompt_version, params (jsonb), latency_ms	exactly one of case_id / vignette_id; references immutable kb_versions
retrieved_references	recommendation_id, rank, chunk_id → kb_chunks, similarity, is_cited	primary key (recommendation_id, rank)
staff_decisions	id, case_id, type, final_category, reason, direction, decided_by, decided_at, amends_id	reason required when type = ADJUST; amends_id → staff_decisions
clinical_notes	id, case_id, author_id, text, created_at	—
audit_log	id (bigserial), ts, user_id, role, action, entity_type, entity_id, before (jsonb), after (jsonb)	INSERT and SELECT only for the application role; trigger rejects UPDATE/DELETE (ADR-12)
jobs	id, type, case_id or vignette_id, status, attempts, last_error, updated_at	used for recovery after restart (ADR-08)
presenting_complaints	code (PK), name	seeded with the fixed complaint list (TBD-4)
kb_entries	id, complaint_code, title, content, species, source_title, source_publisher, source_url, access_date, status, approved_by, approved_at	source fields required; approved_by must have can_approve_kb
red_flag_rules	id, entry_id, indicator, min_category, approved_by	used by stages 2 and 6
kb_chunks	id, entry_id, text, embedding vector(d)	HNSW index on embedding (cosine); d depends on the model (TBD-2)
kb_versions, kb_version_entries	id, version_no, published_at; (version_id, entry_id)	rows are never updated after publishing (DR-05)
evaluation_sets, vignettes	id, name, imported_at; id, set_id, species, description, reference_category, relevant_chunk_ids	separate from clinical tables (DR-06)
evaluation_runs, evaluation_results	id, set_id, kb_version_id, model_id, prompt_version, metrics (jsonb); run_id, vignette_id, recommendation_id, stage latencies	—

7.1.7 Source Code Organization

The Git repository is organized so that each architectural layer and pipeline stage has its own module, which keeps changes local and simplifies testing (NFR-24, NFR-25).

PATH	CONTENTS
frontend/src/pages	One folder per screen W-01 to W-10 (Section 7.3)
frontend/src/components	Shared components: VTL badge, AI label, queue table, reference card, dialogs
frontend/src/api	Typed API client generated from the OpenAPI document
backend/app/api	FastAPI routers grouped by resource (auth, cases, users, kb, eval, exports)
backend/app/core	Configuration, security (hashing, sessions, RBAC dependencies), logging
backend/app/services	User, Case, Review, Knowledge Base, Evaluation, and Audit services
backend/app/pipeline	deidentify.py, redflags.py, extraction.py, retrieval.py, generation.py, safety.py, orchestrator.py
backend/app/adapters	llm_provider.py (interface and implementations), embeddings.py
backend/app/models, schemas, repositories	SQLAlchemy models, Pydantic schemas, repository classes
backend/prompts	Versioned prompt templates and JSON schemas
backend/tests	Unit tests per module and integration tests with mock providers
db/migrations	Alembic migration scripts
deploy	docker-compose.yml, Dockerfiles, nginx.conf, backup script
docs	README (installation and access instructions for PD8), user manual, evaluation protocol

7.2 Architecture Decision Log and Design Rationale

7.2.1 Decision Log

Architecture decisions are recorded as Architecture Decision Records (ADRs). Each record states the context, the decision, the alternatives considered, and the consequences. Accepted decisions guide the PD8 prototype; Proposed decisions will be confirmed through prototype benchmarking

and recorded in later revisions. A decision is changed only by adding a new ADR that supersedes the old one, so the log also serves as the design history.

ID	DECISION	STATUS
ADR-01	Use a layered modular monolith	Accepted
ADR-02	Build the client as a React single-page application	Accepted
ADR-03	Use Python 3.11 and FastAPI for the backend	Accepted
ADR-04	Store relational and vector data in PostgreSQL with pgvector	Accepted
ADR-05	Use retrieval-augmented generation instead of fine-tuning	Accepted
ADR-06	Access the LLM through a provider adapter; hosted API by default	Accepted (pattern); model selection Proposed (TBD-1)
ADR-07	Run the embedding model locally on the server	Proposed (candidate models to be benchmarked; TBD-2)
ADR-08	Process cases asynchronously with an in-process job runner and polling	Accepted
ADR-09	Surround the LLM with deterministic safety controls	Accepted
ADR-10	De-identify text before any external call	Accepted
ADR-11	Use cookie-based sessions with server-side role checks	Accepted
ADR-12	Enforce the append-only audit log in the database	Accepted
ADR-13	Deploy with Docker Compose on one Linux virtual machine	Accepted (hosting provider TBD-3)
ADR-14	Make AI outputs reproducible and traceable	Accepted
ADR-15	Use a three-panel review layout with explicit AI labeling	Accepted

7.2.2 Architecture Decision Records

ADR-01: Use a layered modular monolith

Status	Accepted
Context	The system must be built and deployed by a three-person team within one semester (SRS 2.5) while remaining maintainable (NFR-24).
Decision	Implement one backend service with clearly separated layers (API, services, pipeline, adapters, data access) and one module per domain area.
Alternatives considered	Microservices (one service per process); serverless functions.

Consequences	One deployable unit, simple local development and debugging, and ordinary database transactions. All modules share one runtime and scale together, which is acceptable for up to 10 concurrent users (NFR-06). Module boundaries allow a service to be extracted later if needed.
Related requirements	NFR-06, NFR-24, NFR-26

ADR-02: Build the client as a React single-page application

Status	Accepted
Context	The Triage Review Dashboard is highly interactive: live queue updates, editable extraction, dialogs, and polling (IR-22).
Decision	Use React 18 with TypeScript and Vite; generate a typed API client from the OpenAPI document.
Alternatives considered	Server-rendered templates (e.g., Jinja); other SPA frameworks such as Vue.
Consequences	Reusable components (such as the VTL badge) ensure consistent presentation (IR-03), and typing reduces integration errors. The build tool chain adds some setup effort, and accessibility must be tested explicitly (NFR-21).
Related requirements	IR-01–IR-08, IR-22

ADR-03: Use Python 3.11 and FastAPI for the backend

Status	Accepted
Context	The backend hosts NLP and retrieval code and must validate structured AI output and document its API.
Decision	Implement the backend in Python 3.11 with FastAPI and Pydantic.
Alternatives considered	Django; Node.js with Express.
Consequences	Direct access to the Python NLP ecosystem (embeddings, evaluation metrics), schema validation of LLM output through Pydantic (IR-13), asynchronous calls to external services, and automatically generated OpenAPI documentation (IR-11). FastAPI has no built-in administration interface, so administration screens are built in the SPA.
Related requirements	IR-11, IR-13, FR-66

ADR-04: Store relational and vector data in PostgreSQL with pgvector

Status	Accepted
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Context	Recommendations must reference exact knowledge base versions and passages (FR-26, FR-54), and the knowledge base is small (at most 5,000 chunks, NFR-02).
Decision	Use a single PostgreSQL 16 database with the pgvector extension and an HNSW index for similarity search.
Alternatives considered	A separate vector database; a file-based vector index; SQLite.
Consequences	Foreign keys link recommendations, references, chunks, and versions with full integrity; one backup covers all data; retrieval easily meets the 1-second p95 target at this size. A specialized vector database would scale further, which is not required.
Related requirements	DR-01, DR-05, FR-54, NFR-02, SR-15

ADR-05: Use retrieval-augmented generation instead of fine-tuning

Status	Accepted
Context	No labeled clinical dataset is available (SRS 2.5), and every recommendation must be explainable with references (NFR-18).
Decision	Use a pretrained transformer LLM with prompting and retrieval from the curated knowledge base; do not fine-tune.
Alternatives considered	Fine-tuning a classifier on case data; a purely rule-based classifier.
Consequences	Recommendations are grounded in veterinarian-approved passages and can cite them; knowledge can be updated by approving entries without retraining. Quality depends on retrieval, which is therefore measured separately (Recall@5, MRR; NFR-17).
Related requirements	FR-19–FR-22, NFR-12, NFR-17, NFR-18

ADR-06: Access the LLM through a provider adapter; hosted API by default

Status	Accepted (pattern); model selection Proposed (TBD-1)
Context	The final model is not yet selected, and provider pricing or availability may change during the project.
Decision	Define an LLMProvider interface. Use a hosted LLM API as the default implementation and keep a locally served open-source model as a fallback. Select the model in the prototype by comparing JSON output validity, latency against NFR-01, cost against the budget, and data-use terms (no training on submitted data).
Alternatives considered	Calling one vendor SDK directly from the pipeline; local model only.

Consequences	The model can be changed through configuration without code changes to the pipeline (IR-12), and tests can use a mock provider. The system depends on network access, which is covered by the manual-triage fallback (NFR-09).
Related requirements	IR-12, IR-18, IR-19, NFR-01, NFR-09, NFR-28

ADR-07: Run the embedding model locally on the server

Status	Proposed (candidate models to be benchmarked; TBD-2)
Context	Embeddings are needed for every query and every approved knowledge base chunk.
Decision	Host an open-source sentence-embedding model on the CPU inside the backend container, behind the EmbeddingProvider interface. Choose the model with the highest Recall@5 on the relevance-labeled vignettes that fits the server memory.
Alternatives considered	A hosted embedding API.
Consequences	No per-call cost, no knowledge base or query text leaves the server for embedding, and results are deterministic. The container image is larger and needs about 1–2 GB of additional memory. In the PD6 diagrams, this component is the Embedding Service.
Related requirements	NFR-02, NFR-23, NFR-28, SR-08

ADR-08: Process cases asynchronously with an in-process job runner and polling

Status	Accepted
Context	A pipeline run takes several seconds (two LLM calls), but the interface must stay responsive (NFR-03, NFR-04).
Decision	Return HTTP 202 on submission, run the pipeline as a background job inside the backend, persist job state in the jobs table, and let the client poll for status at most every 15 seconds. On startup, re-queue jobs left in Processing.
Alternatives considered	Synchronous processing within the request; a message broker with worker processes; WebSockets.
Consequences	No additional infrastructure is required and the user can continue working. Polling adds small, regular requests. If load grows, the job runner can be replaced by a broker without changing the API.
Related requirements	FR-06, IR-22, NFR-01, NFR-03, NFR-09

ADR-09: Surround the LLM with deterministic safety controls

Status	Accepted
Context	LLM output can be wrong or unsupported, and under-triage can harm patients (SRS 5.2).
Decision	Apply defense in depth: a keyword red-flag pre-screen before the LLM, citation validation and a veterinarian-approved minimum category after the LLM, conservative tie-breaking, and mandatory staff confirmation.
Alternatives considered	Relying on prompt instructions alone.
Consequences	Critical presentations are flagged within 2 seconds even when the LLM is slow, and safety rules can be tested independently of the model. The red-flag list and rules must be maintained and approved by a veterinarian (BR-04).
Related requirements	FR-12, FR-22, FR-23, FR-25, NFR-05, NFR-07, NFR-08, NFR-12

ADR-10: De-identify text before any external call

Status	Accepted
Context	Owner information is personal data under the Data Privacy Act of 2012, and the LLM is an external service.
Decision	Store owner identifiers in a separate table and remove them, together with pattern-matched phone numbers, e-mail addresses, and street addresses, from the text before stage 3. The intake form instructs staff not to type owner details in the description.
Alternatives considered	Sending the full text and relying on the provider's privacy policy.
Consequences	Only the minimum necessary data leaves the system (SR-08). Rule-based de-identification may miss unusual identifiers, so it is covered by dedicated test cases and user guidance.
Related requirements	IR-20, SR-08, SR-13, DR-04

ADR-11: Use cookie-based sessions with server-side role checks

Status	Accepted
Context	Authentication must resist common web attacks and enforce roles on the server (SR-04, SR-05).
Decision	Issue a signed session token in a Secure, HttpOnly, SameSite cookie with a 30-minute idle expiry; hash passwords with Argon2id; enforce the role matrix with a FastAPI dependency on every endpoint; protect state-changing requests with a CSRF token.

Alternatives considered	Tokens stored in browser localStorage; an external identity provider.
Consequences	Tokens cannot be read by page scripts, which limits the impact of cross-site scripting, and authorization cannot be bypassed through the interface. CSRF protection must be implemented.
Related requirements	FR-57–FR-62, SR-01–SR-05, SR-09

ADR-12: Enforce the append-only audit log in the database

Status	Accepted
Context	Audit entries must never be modified or deleted (FR-43).
Decision	Grant the application database role only INSERT and SELECT on audit_log and add a trigger that rejects UPDATE and DELETE.
Alternatives considered	Enforcement in application code only.
Consequences	The rule holds even if the application has a defect. Corrections are recorded as new entries.
Related requirements	FR-41–FR-43, FR-48, SR-12

ADR-13: Deploy with Docker Compose on one Linux virtual machine

Status	Accepted (hosting provider TBD-3)
Context	The system is needed only for development and the evaluation period and must be deployable by following the README (NFR-26).
Decision	Package the proxy, backend, database, and backup job as containers and deploy them with one Docker Compose command behind an Nginx TLS proxy.
Alternatives considered	A platform-as-a-service offering; Kubernetes.
Consequences	Low cost, identical development and evaluation environments, and simple installation instructions for PD8. The single server is a single point of failure, which is acceptable for the 95% availability target during scheduled sessions (NFR-22) and is mitigated by daily backups.
Related requirements	NFR-22, NFR-26, SR-15, DR-07

ADR-14: Make AI outputs reproducible and traceable

Status	Accepted
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Context	Evaluation results must be repeatable (NFR-23) and each recommendation traceable (FR-26).
Decision	Keep prompt templates as versioned files, use temperature 0, publish immutable knowledge base versions, and store the model identifier, prompt version, parameters, and KB version with every result.
Alternatives considered	Editing prompts in place; storing only the final category.
Consequences	Any result can be explained and re-run with the same configuration. Every prompt change requires a new version and a new evaluation run.
Related requirements	FR-26, FR-54, NFR-23

ADR-15: Use a three-panel review layout with explicit AI labeling

Status	Accepted
Context	Reviewers must verify AI output quickly and must never mistake a recommendation for a decision (NFR-07, NFR-20).
Decision	Show the original description, the extracted information, and the recommendation side by side; label AI content until it is confirmed; always show the VTL color with its name and target time; place decision actions in a fixed bar; follow WCAG 2.1 AA.
Alternatives considered	A single-column form; color-only category indicators.
Consequences	All evidence needed for a decision is visible at once, and the design does not rely on color perception (NFR-21). The three panels need at least 1280 px; on tablets the panels stack vertically.
Related requirements	IR-01–IR-08, FR-31, NFR-07, NFR-20, NFR-21

7.2.3 Design Principles Applied

The following software design principles guided the architecture and detailed design:

PRINCIPLE	APPLICATION IN TRIAGEAI	WHERE
Separation of concerns (layering)	Presentation, API, services, pipeline, adapters, and data access are separate layers with one-way dependencies.	Figure 7.1; ADR-01
High cohesion and low coupling	Each pipeline stage and service has one purpose and exchanges typed data objects instead of sharing state.	Section 7.1.4

PRINCIPLE	APPLICATION IN TRIAGEAI	WHERE
Dependency inversion and open/closed	The pipeline depends on the LLMPProvider and EmbeddingProvider interfaces; new providers are added without modifying the pipeline.	Figure 7.3; ADR-06, ADR-07
Information hiding	Repositories hide SQL and adapters hide provider APIs from the business logic.	Sections 7.1.2, 7.1.7
Defense in depth	Deterministic safety checks, citation validation, Human-in-the-Loop confirmation, and the audit trail protect against AI errors.	ADR-09, ADR-15
Fail-safe defaults	Any AI failure routes the case to manual triage; ties choose the more urgent category; de-identification errors stop external calls.	Section 7.1.4; NFR-09, FR-25
Least privilege	Role checks on every endpoint; the database role cannot alter audit entries or original descriptions; secrets stay on the server.	ADR-11, ADR-12
Privacy by design and data minimization	Owner identifiers are stored separately and never sent to external services.	ADR-10
Design for testability	Deterministic settings, versioned prompts, and mockable interfaces allow automated tests of every stage.	ADR-14; NFR-25
Keep it simple (KISS/YAGNI)	A modular monolith, one database, and an in-process job runner avoid infrastructure the MVP does not need.	ADR-01, ADR-04, ADR-08
Human-centered design	The review screen presents all evidence at once and makes the reviewer's authority explicit.	ADR-15; Section 7.3

7.2.4 Quality Attribute Traceability

The table below shows how the architecture addresses the nonfunctional requirements of the SRS.

QUALITY ATTRIBUTE	ARCHITECTURAL TACTICS	DECISIONS
Safety (NFR-07–NFR-15)	Human-in-the-Loop confirmation, red-flag pre-screen and safety floor, citation validation, fail-safe manual triage, under-triage reporting.	ADR-09, ADR-15
Performance (NFR-01–NFR-06)	Asynchronous jobs with polling, local embeddings, HNSW vector index, stage latency recording.	ADR-04, ADR-07, ADR-08
Security and privacy (SR-01–SR-16)	TLS proxy as the only public entry, cookie sessions, server-side RBAC, de-identification, database privileges, secrets in environment variables.	ADR-10–ADR-13

QUALITY ATTRIBUTE	ARCHITECTURAL TACTICS	DECISIONS
Transparency (NFR-18)	Stored rationale and retrieved references with citation flags for every recommendation.	ADR-05, ADR-09
Reproducibility (NFR-23)	Versioned prompts, temperature 0, immutable KB versions, recorded configuration.	ADR-14
Maintainability and testability (NFR-24, NFR-25)	Layered modular monolith, adapters, repositories, per-module tests with mock providers.	ADR-01, ADR-03, ADR-06
Portability (NFR-26)	Containerized deployment with one Docker Compose command.	ADR-13
Usability and accessibility (NFR-19–NFR-21)	Three-panel layout, consistent components, color plus text, keyboard operation, WCAG 2.1 AA.	ADR-02, ADR-15
Cost and sustainability (NFR-28)	Results stored and reused; LLM called again only after edits; no paid embedding calls.	ADR-07, ADR-08

7.3 User-Interface Wireframes and Prototype Screens

This section presents mid-fidelity wireframes of the main screens. They define the layout, content, and behavior of each screen; the final visual styling will be refined during the PD8 implementation and the usability evaluation. The wireframes were produced as a clickable HTML prototype (TriageAI_Prototype.html, provided with this deliverable), in which the navigation links, queue rows, and main buttons open the related screens. Numbered black markers on each screen are explained in the annotation table that follows it. All names and case data are fictitious sample data.

7.3.1 UI Design Conventions

ELEMENT	CONVENTION	REQUIREMENT
Layout	Designed at 1280 × 800 px for desktops; responsive down to 768 px, where tables become cards and panels stack (W-11).	IR-01
Persistent header	System name, role-based navigation, user name and role, Help, and Log out on every screen after login.	IR-02
VTL badges	Red, Orange, Yellow (dark text), Green, and Blue badges always include the category name and target time; cases without a category use a dashed “MANUAL” badge.	IR-03, NFR-21
AI labeling	AI output is shown with a dashed chip reading “AI Recommendation – requires staff confirmation” until a reviewer decides.	IR-04, NFR-07
Alerts	Red-flag banners use a red border and an exclamation icon; warnings (AI unavailable, safety floor) use a yellow border.	FR-12, NFR-09

ELEMENT	CONVENTION	REQUIREMENT
Errors	Field-level messages in plain language directly below the field; entered data is kept.	IR-05, FR-04
Actions	One primary (filled) button per screen; irreversible actions ask for confirmation; decision actions sit in a fixed bottom bar.	IR-08
Disclaimer	Every recommendation screen shows that TriageAI is decision support and not a diagnosis.	IR-06, NFR-10
Accessibility	Minimum 4.5:1 text contrast, visible keyboard focus, labeled form fields, no information by color alone.	IR-07, NFR-21
Language and formats	English interface text; times in Philippine Standard Time; weight in kilograms.	NFR-27, DR-03

7.3.2 Screen Inventory and Navigation

ID	SCREEN	USERS	USE CASES
W-01	Login	All users	UC-01
W-02	Triage Queue (home)	Intake Staff, Veterinary Reviewer	UC-04
W-03	Case Intake Form	Intake Staff, Veterinary Reviewer	UC-02
W-04	Case Review (Triage Review Dashboard)	Veterinary Reviewer (Intake Staff: view)	UC-05, UC-06, UC-09
W-05	Adjust Urgency Category dialog	Veterinary Reviewer	UC-07
W-06	Case Review – AI unavailable / manual triage	Veterinary Reviewer	UC-08
W-07	Case History and Audit Timeline	Intake Staff, Veterinary Reviewer, Administrator	UC-11
W-08	Knowledge Base Management	Administrator, Veterinary Domain Reviewer	UC-13, UC-14
W-09	User Management	Administrator	UC-12
W-10	Evaluation Console	Administrator	UC-15, UC-16
W-11	Triage Queue – tablet layout	Intake Staff, Veterinary Reviewer	UC-04

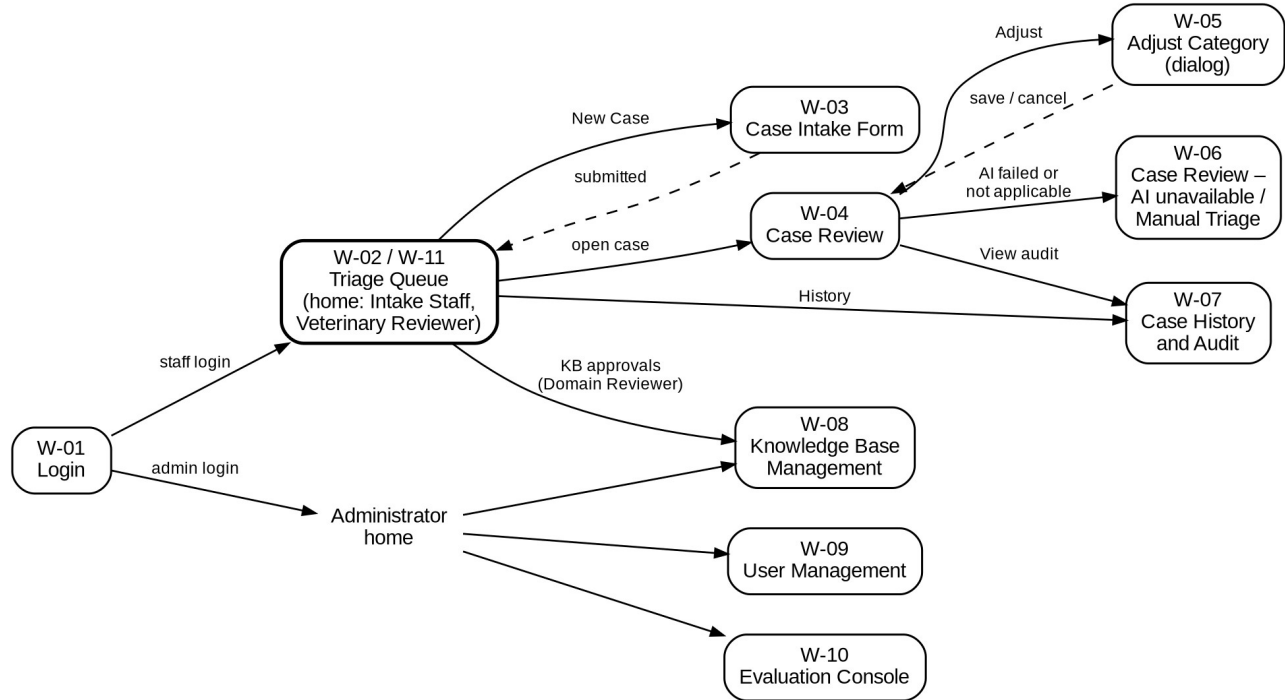


Figure 7.4: Screen Navigation Map

7.3.3 Wireframes

W-01: Login. The login screen authenticates all users. Error messages do not reveal whether the username or the password was wrong.

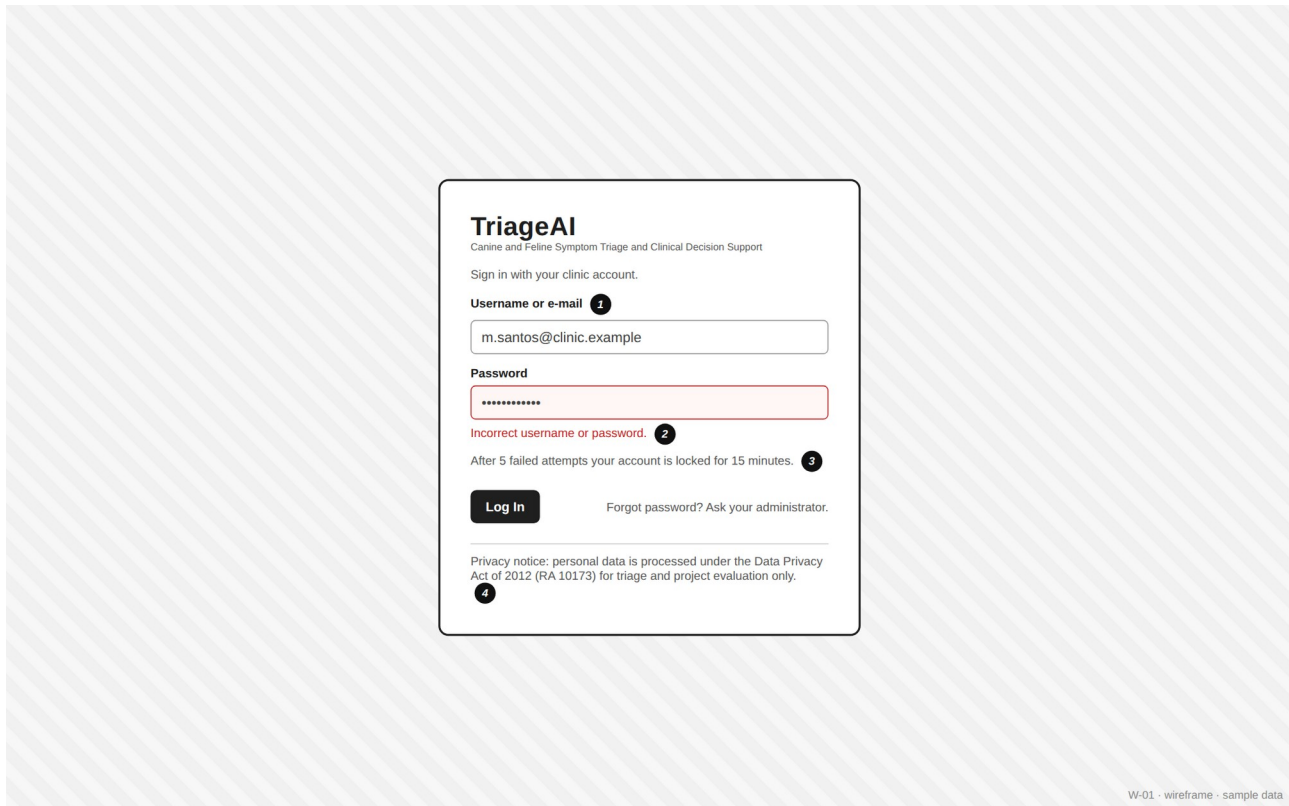
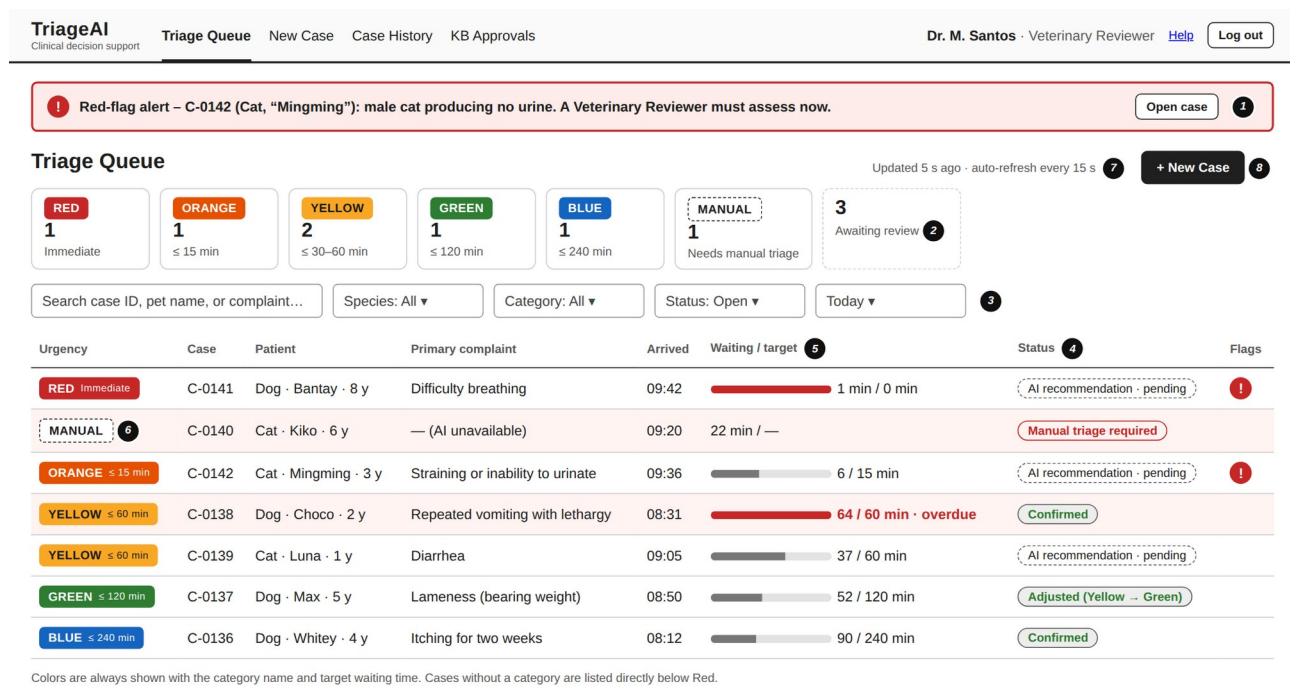


Figure 7.5: W-01 Login

NO.	ELEMENT AND BEHAVIOR	REQUIREMENTS
1	Username/e-mail and password fields	FR-57
2	Generic error message after a failed attempt	SR-02, IR-05
3	Lockout rule shown in advance: 15 minutes after 5 failed attempts	SR-03
4	Privacy notice under the Data Privacy Act of 2012	SR-13

W-02: Triage Queue. The Triage Queue is the home screen for clinic staff. Open cases are sorted by urgency and then arrival time, and cases without a category are shown directly below Red so they are not overlooked.



W-02 - wireframe - sample data

Figure 7.6: W-02 Triage Queue

NO.	ELEMENT AND BEHAVIOR	REQUIREMENTS
1	Red-flag banner stays visible until the case is opened and links directly to it	FR-12, BR-02, NFR-05
2	Counts per category with names and target times, plus cases awaiting review	IR-03, FR-29
3	Search and filters by species, category, status, and date	FR-39, FR-44
4	Status chips distinguish AI recommendations pending review (dashed) from staff decisions	FR-36, IR-04
5	Waiting time against target time; overdue rows are tinted and marked “overdue”	FR-29, FR-30
6	Manual-triage case with a dashed MANUAL badge after an AI failure	NFR-09, FR-34
7	Automatic refresh indicator (polling every 15 s)	IR-22
8	New Case button opens W-03	FR-01

W-03: Case Intake Form. The intake form records the case in under a minute. The description is entered in English (FR-18); the helper text instructs staff to translate Filipino or Bikol wording and to quote uncertain terms. The side panel reminds staff of red-flag signs that require an immediate escalation.

TriageAI
Clinical decision support

[Triage Queue](#)
[New Case](#)
[Case History](#)

J. Cruz · Intake Staff
[Help](#)
[Log out](#)

New Triage Case

Species (required) 1

Dog
Cat
Other species

Other species are not processed by the AI and must be triaged manually.

Pet name (optional) 2

Age
 years

Sex and neuter status

Breed

Body weight (kg)

Enter the weight as a number, e.g. 4.2 5

Intake channel

Walk-in
Phone
Message

Owner's description of the problem (required, 20–2,000 characters) 3

Since this morning he keeps going to the litter box and cries, but nothing comes out. He also vomited twice. He is 3 years old and was fine last night.

Enter the description in English; translate any Filipino or Bikol words, keeping the owner's exact wording in quotes if unsure. Do not include the owner's name or phone number here.
143 / 2,000

Owner reference (optional) 4 – stored separately and never sent to AI services

Owner name

Contact number

Submit for triage 6
Cancel

The recommendation usually appears within 10 seconds.

Red-flag signs – notify a Veterinary Reviewer immediately

- Not breathing or struggling to breathe
- Collapsed or unresponsive
- Seizure happening now
- Bleeding that will not stop
- Male cat straining with no urine
- Possible poison or medication swallowed
- Pale, white, or bluish gums

Final list to be validated by the veterinary reviewer (TBD-4).

Tips for a useful description
Ask when it started, how often it happens, whether it is getting worse, and whether the pet is eating and drinking.

Figure 7.7: W-03 Case Intake Form

NO.	ELEMENT AND BEHAVIOR	REQUIREMENTS
1	Species control; “Other species” is routed to manual triage	FR-02
2	Optional signalment fields	FR-01
3	Owner’s description with length limits, counter, and guidance; stored verbatim	FR-01, FR-05
4	Owner reference in a separate section that is never sent to AI services	FR-07, DR-04, IR-20
5	Field-level validation message; other entries are kept	FR-04, IR-05
6	Submit starts the pipeline; the expected waiting time is shown	FR-06, NFR-01

W-04: Case Review (Triage Review Dashboard). The Case Review screen is the core Human-in-the-Loop interface. The three panels show the evidence, the AI interpretation, and the recommendation side by side, and the decision bar remains visible while scrolling.

TriageAI
Clinical decision support

Triage Queue New Case Case History KB Approvals

Dr. M. Santos · Veterinary Reviewer [Help](#) [Log out](#)

[← Back to queue](#)

C-0142 · Cat · Mingming · 3 y · Male, intact ¹

Arrived 09:36 by phone · recorded by J. Cruz · waiting 6 min of 15 min

AI Recommendation – requires staff confirmation ²

Red flag detected by the pre-screen: male cat producing no urine.

Owner's description (verbatim) ³

Since this morning he keeps going to the litter box and cries, but nothing comes out. He also vomited twice. He is 3 years old and was fine last night.

Highlights show the evidence used for each extracted value.

Extracted information ⁴ Click a value to correct it

Species: Cat

Presenting complaints: Straining or inability to urinate (primary), Vomiting

Onset / duration: Since this morning (< 24 h)

Frequency / severity: Repeated attempts; vomited 2 times

Associated signs: Vocalizing (crying)

Negated findings: None stated

Red flags: Male cat producing no urine

Ask the owner: Appetite · water intake · previous episodes

[Save and regenerate](#) enabled after a correction

Recommended urgency ⁵

ORANGE · Very urgent · see within 15 min

Safety floor applied: the red flag "male cat, no urine" requires at least Orange. ⁶

Rationale. The owner reports repeated straining with no urine since this morning in a male cat, with vomiting. The knowledge base identifies this pattern as a possible urinary blockage that needs prompt veterinary assessment.

Confidence: Medium ■■■ · [View audit trail](#)

References ⁷

[1] [Merck Veterinary Manual](#) – Feline lower urinary tract disease (urethral obstruction section). Summary: male cats that strain without passing urine may have a blockage that requires prompt care. [Open source](#) ↗

[2] [Veterinary Triage List](#) – urinary discriminators (Ruys et al., 2012). Summary: inability to urinate is listed among discriminators for the higher urgency categories. [Open source](#) ↗

Model: TBD-1 · Prompt v1.3 · KB version 7 · generated 09:36:14 in 9.2 s ⁸

[Confirm Orange](#) [Adjust category...](#) [Mark for manual triage](#) [Add note](#) ⁹

Decision support only – this is not a diagnosis. The final urgency category is decided by veterinary staff. ¹⁰

W-04 · wireframe · sample data

Figure 7.8: W-04 Case Review (Triage Review Dashboard)

NO.	ELEMENT AND BEHAVIOR	REQUIREMENTS
1	Case header with signalment and live waiting time against target	FR-29, FR-31
2	AI label until a Veterinary Reviewer confirms or adjusts	IR-04, NFR-07
3	Verbatim description with highlighted evidence spans	FR-13
4	Editable extracted information, missing-information prompts, and Save and regenerate	FR-14, FR-16, FR-27
5	VTL badge with category name and target waiting time	IR-03, FR-20
6	Safety-floor notice when a red-flag rule raised the category	FR-23
7	References with source, section, summary, and link	FR-22, FR-31, NFR-18
8	Confidence indicator and configuration trace (model, prompt, KB version, time)	FR-24, FR-26
9	Decision actions: Confirm, Adjust, Manual triage, Add note (Veterinary Reviewer only)	FR-32–FR-35, FR-37
10	Decision-support disclaimer	IR-06, NFR-10

W-05: Adjust Urgency Category Dialog. The dialog is opened from W-04. It makes the direction of the change explicit and requires a reason before the decision can be saved.

The screenshot shows the 'Adjust urgency category' dialog box in the TriageAI interface. The dialog is titled 'Adjust urgency category' and contains the following elements:

- AI recommendation:** A message indicating the current recommendation is 'ORANGE' and instructing the user to select a category.
- Urgency Categories:** Five radio button options are listed:
 - RED:** Immediate · 0 min (Up-triage)
 - ORANGE:** Very urgent · ≤ 15 min (current)
 - YELLOW:** Urgent · ≤ 30–60 min
 - GREEN:** Standard · ≤ 120 min
 - BLUE:** Non-urgent · ≤ 240 min
- Reason for change (required):** A dropdown menu with the selected reason: 'Patient's condition on arrival differs from the description'.
- Details:** A text field containing the detail: 'On arrival: abdomen painful, cat weak and lying down.'
- Summary:** A message at the bottom stating: 'Change: Orange -> Red (up-triage). Your name, the time, and the reason are recorded in the audit trail and cannot be edited later.'
- Buttons:** 'Cancel' and 'Save decision' buttons are located at the bottom right.

Figure 7.9: W-05 Adjust Urgency Category Dialog

NO.	ELEMENT AND BEHAVIOR	REQUIREMENTS
1	All five categories with target times; the AI recommendation is marked as current	FR-33
2	Required reason from a predefined list, with optional details	FR-33, BR-03
3	Summary of the change (up- or down-triage) and notice that it is recorded permanently	IR-08, FR-42, FR-48

W-06: Case Review – AI Unavailable. When an AI service fails, the case is never blocked or lost: the reviewer sees the reason, can retry, and can triage manually with the VTL reference on the same screen.

TriageAI
Clinical decision support

[Triage Queue](#)
[New Case](#)
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Dr. M. Santos · Veterinary Reviewer
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C-0140 · Cat · Kiko · 6 y · Female, spayed

AI unavailable – triage manually using the VTL. The language model did not respond after two retries (09:20:41). The case is safe to triage; no data was lost.

1 [Retry AI processing](#)

Owner's description (verbatim)
 She has been hiding under the bed since yesterday and did not eat her breakfast. She is breathing a bit fast when I touch her belly.
 Extracted information: not available for this case. 2

Assign a category manually 3

☐ **RED** Immediate · e.g. not breathing, unresponsive, ongoing seizure, uncontrolled bleeding

☒ **ORANGE** Very urgent · e.g. breathing difficulty, severe pain, suspected toxin, male cat unable to urinate

☐ **YELLOW** Urgent · e.g. repeated vomiting with lethargy, controlled bleeding

☐ **GREEN** Standard · e.g. lameness while bearing weight, single vomiting episode

☐ **BLUE** Non-urgent · e.g. chronic itching, mild eye or ear discharge

Note (required)
 Fast breathing reported; assess on arrival.

Save manual triage

 Status will change to "Manually triaged". 4

W-06 - wireframe - sample data

Figure 7.10: W-06 Case Review – AI Unavailable

NO.	ELEMENT AND BEHAVIOR	REQUIREMENTS
1	"AI unavailable – triage manually using the VTL" banner with the reason and a Retry button	NFR-09, IR-19
2	Extraction panel shows that no AI output exists for this case	FR-15
3	Manual category selection with VTL example discriminators	FR-34, NFR-13
4	Saving sets the status to Manually Triaged and requires a note	FR-34, FR-41

W-07: Case History and Audit Timeline. The history screen supports review of past cases and of disagreements between the AI and staff.

TriageAI
Clinical decision support

Triage QueueNew CaseCase History

Dr. M. Santos · Veterinary ReviewerHelpLog out

Case History

Case ID

01–30 Oct 2026 ▼

Species: All ▼

Complaint: All ▼

Category: All ▼

Status: All ▼

Search1

Case	Date	Patient	AI	Final	Status
C-0137	Today 08:50	Dog · Max	YELLOW	GREEN	Closed
C-0136	Today 08:12	Dog · Whitey	BLUE	BLUE	Confirmed
C-0135	Yesterday	Dog · Brownie	ORANGE	RED	Closed
C-0134	Yesterday	Cat · Snow	GREEN	GREEN	Closed
C-0133	Yesterday	Dog · Tiger		YELLOW	Manually triaged

AI and final categories are shown side by side so disagreements are easy to review.2

Audit timeline – C-0137

Submission → recommendation: 10 s

Recommendation → decision: 8 min 28 s

08:50:02 · Case created
J. Cruz (Intake Staff) · channel: walk-in

08:50:03 · Red-flag pre-screen
No red flags found

08:50:09 · Extraction v1 stored
Prompt v1.3 · schema valid

08:50:12 · Recommendation v1: YELLOW
Confidence Medium · KB version 7 · 2 references

08:58:40 · Decision: Adjusted Yellow → Green
Dr. M. Santos · reason: "Bearing weight, alert, no swelling on arrival."

09:40:10 · Note added
Dr. M. Santos

10:05:31 · Case closed
Seen by veterinarian

Entries are append-only and cannot be edited or deleted.

W-07 · wireframe · sample data

Figure 7.11: W-07 Case History and Audit Timeline

NO.	ELEMENT AND BEHAVIOR	REQUIREMENTS
1	Search by case ID, date range, species, complaint, category, and status	FR-44
2	AI and final categories side by side	FR-41, BR-03
3	Chronological, append-only audit timeline with users, roles, and reasons	FR-42, FR-43, FR-45
4	Computed intervals: submission to recommendation and recommendation to decision	FR-46

W-08: Knowledge Base Management. Administrators draft entries and Veterinary Domain Reviewers approve them. The example shows an entry pending review as seen by an approver.

TriageAI
Administration

UsersKnowledge BaseEvaluationExports

Dr. A. Rivera · Veterinary Reviewer (KB approver)Log out

Knowledge Base

Published version: v7 · 18 active entries 1

Draft (3)Pending review (2)Active (18)Retired (1)2

Entry	Species	Source	Updated
Straining or inability to urinate	Cat	Merck Vet Manual	Today
Heat exposure or overheating	Dog, Cat	WSAVA	Yesterday

Straining or inability to urinate – Cat

Pending review

Presenting complaint (fixed list)

Straining to urinate or inability to urinate ▼

Species

Cat ▼

Content (written by the curator from the cited source) 3

Male cats that repeatedly strain in the litter box and pass little or no urine may have a urethral blockage. The condition can worsen within hours and should be assessed promptly...

Red flags and minimum category 4

Male cat producing no urine

ORANGE minimum

Collapsed or unresponsive with urinary signs

RED minimum

Source citation (required) 5

Merck Veterinary Manual

merckvetmanual.com/...

Accessed 24 Sep 2026

Approve and publish (v8)

Reject with comments 6

Visible only to licensed veterinarians with approval permission.

W-08 - wireframe - sample data

Figure 7.12: W-08 Knowledge Base Management

NO.	ELEMENT AND BEHAVIOR	REQUIREMENTS
1	Currently published knowledge base version and number of active entries	FR-54
2	Tabs for the Draft → Pending Review → Active → Retired workflow	FR-51
3	Content written by the curator from the cited source (not copied)	SRS Section 6 (legal)
4	Red-flag rules with their minimum VTL category	FR-49, FR-23
5	Required source citation: title, URL, access date	FR-50
6	Approve and publish / Reject with comments, shown only to approvers	FR-52, FR-53, BR-04

W-09: User Management. The Administrator manages accounts and roles.

TriageAI
Administration

UsersKnowledge BaseEvaluationExports

System Administrator · AdministratorLog out

User Management

Name	E-mail	Role 1	KB approval	Status	Last login	
Dr. M. Santos	m.santos@...	Veterinary Reviewer	—	Active	Today 08:01	Edit
Dr. A. Rivera	a.rivera@...	Veterinary Reviewer	Yes	Active	Yesterday	Edit
J. Cruz	j.cruz@...	Intake Staff	—	Active	Today 07:55	Edit
R. Dela Peña	r.delapena@...	Intake Staff	—	Locked until 10:15 2	Today 09:58	Unlock
T. Garcia	t.garcia@...	Intake Staff	—	Deactivated	12 Sep	Reactivate

Add user 3

Full name

E-mail

Role

Intake Staff ▼

☐ Grant knowledge base approval permission
Only for licensed veterinarians with the Veterinary Reviewer role (BR-04). 4

A temporary password is generated and must be changed at first login.

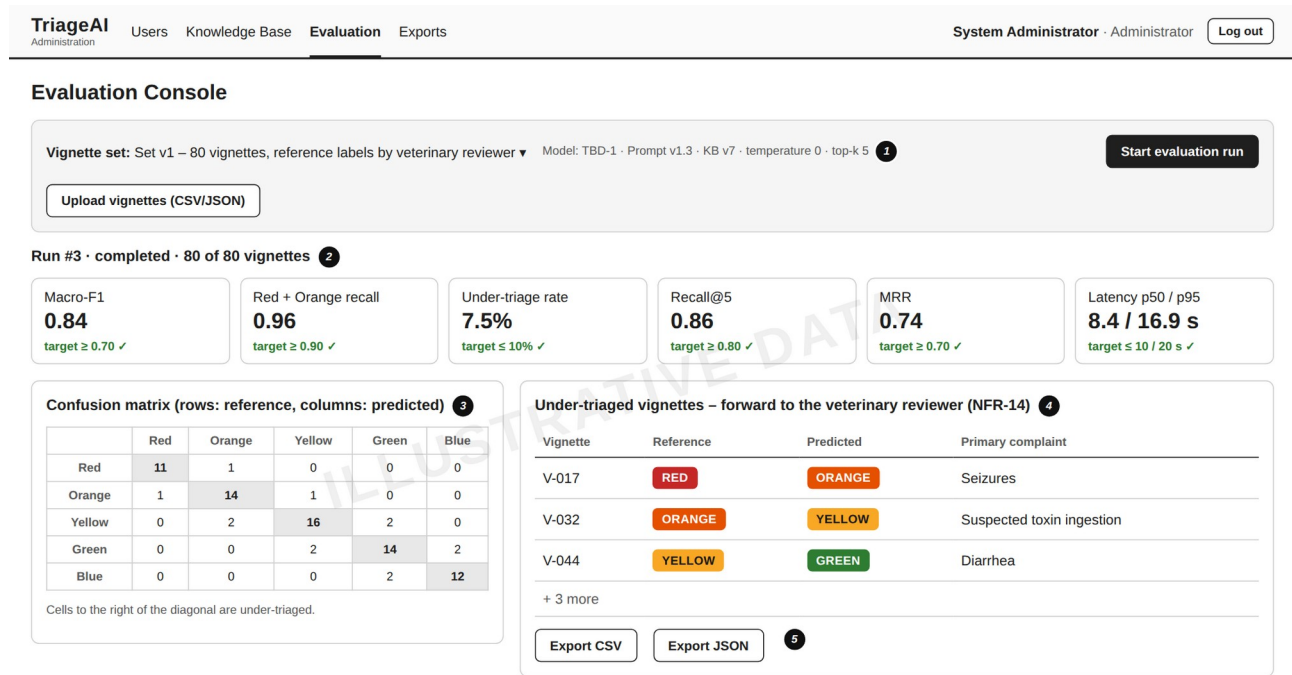
Create account

W-09 - wireframe - sample data

Figure 7.13: W-09 User Management

NO.	ELEMENT AND BEHAVIOR	REQUIREMENTS
1	Role and knowledge base approval permission per user	FR-58
2	Locked account with an Unlock action	SR-03
3	Add user with a temporary password to be changed at first login	FR-60, FR-61
4	Approval permission restricted to licensed veterinarians	FR-58, BR-04

W-10: Evaluation Console. The console runs the vignette set and presents the metrics against the SRS targets. The values shown are illustrative only and are not project results.



W-10 · wireframe · illustrative data, not project results

Figure 7.14: W-10 Evaluation Console

NO.	ELEMENT AND BEHAVIOR	REQUIREMENTS
1	Recorded run configuration: model, prompt version, KB version, parameters	FR-65, NFR-23
2	Metric cards with the SRS targets	FR-66–FR-68, NFR-01, NFR-16, NFR-17
3	Confusion matrix; cells to the right of the diagonal are under-triage	FR-66
4	List of under-triaged vignettes to be reviewed by the veterinarian	NFR-14
5	Export of results as CSV or JSON	FR-69

W-11: Triage Queue – Tablet Layout. At widths below 900 px the header wraps, the category counters form a grid, and each queue row becomes a card with labeled fields, so the queue remains usable on a 768-px tablet. The numbered markers have the same meaning as in W-02.

TriageAI
Clinical decision support

Triage Queue New Case Case History KB Approvals

Dr. M. Santos · Veterinary Reviewer [Help](#) [Log out](#)

Red-flag alert – C-0142 (Cat, “Mingming”): male cat producing no urine. A Veterinary Reviewer must assess now.

[Open case](#) **1**

Triage Queue Updated 5 s ago · auto-refresh every 15 s **7** [+ New Case](#) **8**

RED

1

Immediate

ORANGE

1

≤ 15 min

YELLOW

2

≤ 30–60 min

GREEN

1

≤ 120 min

BLUE

1

≤ 240 min

MANUAL

1

Needs manual triage

3

Awaiting review **2**

Search case ID, pet name, or complaint...

Species: All ▼ Category: All ▼ Status: Open ▼ Today ▼ **3**

Urgency

Case

Patient

Complaint

Arrived

Waiting

Status

Flags

RED · Immediate

C-0141

Dog · Bantay · 8 y

Difficulty breathing

09:42

1 min / 0 min

AI recommendation · pending

!

Urgency

Case

Patient

Complaint

Arrived

Waiting

Status

MANUAL

6

C-0140

Cat · Kiko · 6 y

— (AI unavailable)

09:20

22 min / —

Manual triage required

Figure 7.15: W-11 Triage Queue – Tablet Layout (768 px)

7.3.4 Prototype Use and Next Steps

The clickable prototype will be used for an early walkthrough with the veterinary domain reviewer and one or two clinic staff members before implementation. Their feedback, together with the SUS-based usability evaluation of the PD8 prototype (NFR-19), will be recorded in the revision history.

The screens will then be implemented as React pages following the source code organization in Section 7.1.7.